Coronagraph Merit Function

Reference
Stark *et al.* (2014) *ApJ*

Preamble

Stark *et al.* list four elements in optimizing exoEarth detection:

1. Target list selection
2. Completeness
3. Exposure time
4. Yield maximization

Coronagraph design pertains to only 2 and 3:

- Detection limit (Δmag_{obs}), inner working andgle (IWA) and, to a lesser degree, outer working angle (OWA) determine completeness, $C_i(\Delta mag_{obs}, IWA, OWA)$.
- Exposure time (τ) depends on the following coronagraphic metrics:
  - Υ: the fraction of the diffraction-limited PSF contained within the aperture defined by the angular radius $\theta = X\lambda/D$. Stark *et al.* assume simple aperture photometry with $X = 0.7$ such that $\Upsilon = 0.69$, which roughly maximizes the planet-to-background flux ratio.
  - T: throughput (P. Chen note: throughput is the overall transmittance, ratio of photons on the camera to photons on the entrance pupil for the targeted object)
  - $\Delta\lambda$: wavelength badnwidth
  - ζ: uniform contrast level of suppressed starlight measured relative to the PSF peak per unit solid angle, $PSF_{peak} = \pi D^2/(4\lambda^2)$ expresses the theoretical Airy pattern peak per unit solid angle under the assumption of a diffraction-limited PSF, and $\Omega = \pi(X\lambda/D)^2$ is the solid angle subtended by the photometric aperture.

Completeness is not described by an analytical formula, so we focus on exposure time in deriving a merit function and devise *ad hoc* ways to account for completeness.

Relevant Equations from Stark *et al.*

Required Exposure Time

$$\tau = (S/N)^2 \left(\frac{CR_p + 2CR_b}{CR_p^2} \right)$$

- CR_p is the photon count rate for the planet
- CR_b is the photon count rate for the background
- Factor of two in fron of CR_b is due to the necessity of background subtraction

Planet Photon Rate

$$CR_p = F_0 10^{-0.4(m_V + \Delta mag_{obs})} \frac{\pi D^2}{4} \Upsilon T \Delta\lambda$$

- F_0 is zero-magnitude flux at V band
- Δmag_{obs}: contrast magnitude representing the faintest planet that can be detected at the desired S/N in a certain exposure time.

Background Photon Rate

Leaked Stellar Light

$$CR_{b,\star} = F_0 10^{-0.4 m_V} \zeta \frac{\pi^2 X^2}{4} \frac{\pi D^2}{4} T \Delta\lambda$$

Local Zodiacal Light

$$CR_{b,zodi} = F_0 10^{-0.4z} \frac{\pi^2 X^2}{4(4.848 \times 10^{-6})^2} \lambda_V^2 T \Delta\lambda$$

- z is the typical surface brightness of the zodiacal light in magnitudes per unit solid angle at V band. +
- $(4.848 \times 10^{-6})^2$ is the number of radians in one arcsecond.

Exozodiacal Light

$$CR_{b,exozodi} = F_0 n 10^{-0.4x} \frac{\pi^2 X^2}{4(4.848 \times 10^{-6})^2} \lambda_V^2 T \Delta\lambda$$

- x is the surface brightness of 1 zodi of exozodiacal light in magnitudes per unit solid angle
- n is the typical number of zodis assumed for all stars

Combine the above Equations

$$\tau = (S/N)^2 \left(\frac{1 + 2(CR_b/CR_p)}{CR_p} \right)$$

$$\frac{CR_b}{CR_p} = \frac{F_0 10^{-0.4 m_V} \zeta \frac{\pi^2 X^2}{4} \frac{\pi D^2}{4} T \Delta\lambda + F_0 10^{-0.4 z} \frac{\pi^2 X^2}{4(4.848\times10^{-6})^2} \lambda_V^2 T \Delta\lambda + F_0 n 10^{-0.4 x} \frac{\pi^2 X^2}{4(4.848\times10^{-6})^2} \lambda_V^2 T \Delta\lambda}{F_0 10^{-0.4(m_V+\Delta mag_{obs})} \frac{\pi D^2}{4} \Upsilon T \Delta\lambda}$$

$$\frac{CR_b}{CR_p} = \frac{\frac{\pi^2 X^2}{4}\left[10^{-0.4 m_V} \zeta \frac{\pi D^2}{4} + 10^{-0.4(z-26.57)} \lambda_V^2 + n 10^{-0.4(x-26.57)} \lambda_V^2\right]}{10^{-0.4(m_V+\Delta mag_{obs})} \frac{\pi D^2}{4} \Upsilon}$$

$$\tau = (S/N)^2 \left(\frac{1 + 2\left(\frac{\frac{\pi^2 X^2}{4}\left[10^{-0.4 m_V} \zeta \frac{\pi D^2}{4} + 10^{-0.4(z-26.57)} \lambda_V^2 + n 10^{-0.4(x-26.57)} \lambda_V^2\right]}{10^{-0.4(m_V+\Delta mag_{obs})} \frac{\pi D^2}{4} \Upsilon} \right)}{F_0 10^{-0.4(m_V+\Delta mag_{obs})} \frac{\pi D^2}{4} \Upsilon T \Delta\lambda} \right)$$

$$\tau = (S/N)^2 \left(\frac{10^{-0.4(m_V+\Delta mag_{obs})} \frac{\pi D^2}{4} \Upsilon + 2\left(\frac{\pi^2 X^2}{4}\left[10^{-0.4 m_V} \zeta \frac{\pi D^2}{4} + 10^{-0.4(z-26.57)} \lambda_V^2 + n 10^{-0.4(x-26.57)} \lambda_V^2\right]\right)}{\left(10^{-0.4(m_V+\Delta mag_{obs})} \frac{\pi D^2}{4} \Upsilon\right)\left(F_0 10^{-0.4(m_V+\Delta mag_{obs})} \frac{\pi D^2}{4} \Upsilon T \Delta\lambda\right)} \right)$$

$$\tau = (S/N)^2 \left(\frac{10^{-0.4(m_V+\Delta mag_{obs})} \frac{\pi D^2}{4} \Upsilon + 2\left(\frac{\pi^2 X^2}{4}\left[10^{-0.4 m_V} \zeta \frac{\pi D^2}{4} + 10^{-0.4(z-26.57)} \lambda_V^2 + n 10^{-0.4(x-26.57)} \lambda_V^2\right]\right)}{\left(10^{-0.4(m_V+\Delta mag_{obs})} \frac{\pi D^2}{4} \Upsilon\right)^2 (F_0 T \Delta\lambda)} \right)$$

Merit Function

Maximizing the following function minimizes exposure time.

$$\mathcal{M} = \frac{\left(10^{-0.4(m_V+\Delta mag_{obs})} \frac{\pi D^2}{4} \Upsilon\right)^2 (F_0 T \Delta\lambda)}{10^{-0.4(m_V+\Delta mag_{obs})} \frac{\pi D^2}{4} \Upsilon + 2\left(\frac{\pi^2 X^2}{4}\left[10^{-0.4 m_V} \zeta \frac{\pi D^2}{4} + 10^{-0.4(z-26.57)} \lambda_V^2 + n 10^{-0.4(x-26.57)} \lambda_V^2\right]\right)}$$

$$\mathcal{M} = \frac{F_0 B_1^2 \Upsilon^2 T \Delta\lambda}{B_1 \Upsilon + B_2 B_3 \zeta + B_2 B_4}$$

- $B_1 \equiv 10^{-0.4(m_V + \Delta mag_{obs})} \left(\frac{\pi D^2}{4}\right)$
- $B_2 \equiv 2\left(\frac{\pi^2 X^2}{4}\right)$
- $B_3 \equiv 10^{-0.4 m_V} \left(\frac{\pi D^2}{4}\right)$
- $B_4 \equiv \left[10^{-0.4(z-26.57)} + n10^{-0.4(x-26.57)}\right] \lambda_V^2$
- Use weighting w.r.t. angular position to account for completeness

In [1]:
```
import numpy as np
```

In [2]:
```
F_0 = 9500e13
m_v = 5.0
delta_mag = 25
D = 8.0
encircled_psf = 0.69
z = 23
x = 22
n = 3
X = 0.7
wvl = 550e-9
throughput = 0.2
wvl_bw = 0.11e-6
contrast = 1e-10
SNR = 7.0
```

In [3]:
```
CR_p = F_0*10**(-0.4*(m_v+delta_mag))*(np.pi*D**2/4.0)*encircled_psf*throughput
*wvl_bw
```

In [4]:
```
print("planet count rate:  {:.2E} photons/s".format(CR_p))
```

```
planet count rate:  7.25E-02 photons/s
```

In [5]:
```
CR_b_star = F_0*10**(-0.4*m_v)*contrast*((np.pi**2)*(X**2)/4.0)*(np.pi*D**2/4.
0)*throughput*wvl_bw
```

In [6]:
```
print("residual stellar count rate:  {:.2E} photons/s".format(CR_b_star))
```

```
residual stellar count rate:  1.27E-01 photons/s
```

In [7]:
```
CR_b_zodi = F_0*10**(-0.4*z)*(np.pi**2)*(X**2)/(4.0*np.radians(1/3600.0)**2)*wv
l**2*throughput*wvl_bw
```

In [8]:
```
print("local zodiacal count rate:  {:.2E} photons/s".format(CR_b_zodi))
```

```
local zodiacal count rate:  2.05E-02 photons/s
```

In [9]:
```
CR_b_exozodi = F_0*n*10**(-0.4*x)*(np.pi**2)*(X**2)/(4.0*np.radians(1/3600.0)**
2)*wvl**2*throughput*wvl_bw
```

In [10]:
```
print("exozodiacal count rate:  {:.2E} photons/s".format(CR_b_exozodi))
```

```
exozodiacal count rate:  1.55E-01 photons/s
```

In [11]:
```
CR_b = CR_b_star + CR_b_zodi + CR_b_exozodi
```

In [12]:
```
print("total background count rate:  {:.2E} photons/s".format(CR_b))
```

```
total background count rate:  3.02E-01 photons/s
```

In [13]:
```
tau = SNR**2*(CR_p+2*CR_b)/CR_p**2
```

In [14]:
```
print("required exposure time:  {:.2E} s".format(tau))
```

```
required exposure time:  6.31E+03 s
```

In [15]:
```
parsec = 2.06265e5  # in AU
a_venus = 0.723  # Venus' semi-major axis in AU
a_mars = 1.524

print("{:.2f} m diameter telescope".format(D))
print("Venus angular position (Airy units) @ 10 parsec:  {:.2f}".format((a_venu
s/(10*parsec)*(D/wvl))))
print("Earth angular position (Airy units) @ 10 parsec:  {:.2f}".format((1.0/(1
0*parsec)*(D/wvl))))
print("Mars angular position (Airy units) @ 10 parsec:  {:.2f}".format((a_mars
/(10*parsec)*(D/wvl))))
```

```
8.00 m diameter telescope
Venus angular position (Airy units) @ 10 parsec:  5.10
Earth angular position (Airy units) @ 10 parsec:  7.05
Mars angular position (Airy units) @ 10 parsec:  10.75
```

In [16]:
```
D = 4.0
parsec = 2.06265e5  # in AU
a_venus = 0.723  # Venus' semi-major axis in AU
a_mars = 1.524

print("{:.2f} m diameter telescope".format(D))
print("Venus angular position (Airy units) @ 10 parsec:  {:.2f}".format((a_venu
s/(10*parsec)*(D/wvl))))
print("Earth angular position (Airy units) @ 10 parsec:  {:.2f}".format((1.0/(1
0*parsec)*(D/wvl))))
print("Mars angular position (Airy units) @ 10 parsec:  {:.2f}".format((a_mars
/(10*parsec)*(D/wvl))))
```

```
4.00 m diameter telescope
Venus angular position (Airy units) @ 10 parsec:  2.55
Earth angular position (Airy units) @ 10 parsec:  3.53
Mars angular position (Airy units) @ 10 parsec:  5.37
```

In [ ]:
```

```